

Homework 4 (20pts)

Due December 4 at 11:59 PM Eastern Time

Why are we doing this? The homework assignments are a low-stakes way to practice the concepts we've talked about in class and apply them to real-world problems. For this particular assignment, I would like you to (1) use generative networks test and (2) compare different vaccination strategies.

How will you be assessed? I will run your code to check if the produced results follow the correct trend. Because the model we will use for the simulation is stochastic, the results will not be exact but still follow a general trend. The answers comparing the strategies need not be correct, but should follow a logic which you can (and will have to) explain.

What will you submit? A PDF of the homework assignment and all of the code (Jupyter notebooks or scripts) used to generate the results and figures to the Canvas assignment.

Format requirements

- The title of the homework assignment
- Your name
- The collaborators with whom you worked
- Each problem should have a separate heading
- Plots should have descriptive axis labels
- OPTIONAL helpful tips for plots: (1) it's nice when legends and axis labels follow the format "Descriptive text, symbol" (For example, you would label an axis something like, "Contagion rate, β " by writing the following in Python: `plt.xlabel(r'Contagion rate, $\beta$')`), (2) when saving figures (with `savefig`), either save in vector graphic format (SVG, PDF) or use the `dpi` option and set it to 300 or more (I like 1,000 DPI), (3) use sufficiently large fonts, (4) a legend when multiple things are being plotted on the same plot, (5) use the `despine` command in Seaborn to remove the box around plots, and (6) the `tight_layout` command can automatically adjust the plot spacing to make enough space around the labels, etc.

The assignment

Extra credit (1 point): Use the LaTeX template (See Canvas) to write up the homework.

Question 1 (20 points): In this problem, I want you to imagine that you are a policymaker with the future of the world (a generative network) in your hands. An epidemic has broken out and you need to handle it! You will be testing different vaccination strategies and comparing them against each other to see which is the best. The simulation should go as follows:

- i) Create your world! This will just be a `barabasi_albert_graph` with `n = 100` and `m = 2`.
- ii) You will evaluate the performance of each vaccination strategy across a range of vaccination coverage. Use `np.linspace(0.1, 1, 101)` to loop over each proportion.
- iii) Create arrays or lists to record the outcome of each simulation for every vaccination fraction.
- iv) Now you can run your loop! For each vaccination fraction in your predefined range, run 50 independent realizations of the model to reduce stochastic noise. Store the average result of

these 50 runs in your storage variables.

- v) For each realization, determine the set of vaccinated individuals based on the following strategies:
 - a) No vaccination: So just an empty list, to run as a control.
 - b) Random Vaccination: Randomly choose a fraction of nodes to have been vaccinated.
 - c) Eigenvector Vaccination: Choose nodes with the weighted probability of their eigenvector centrality.
 - d) Degree Vaccination: Choose nodes with the weighted probability of their degree.
 - e) Nomination Vaccination: Randomly select a set of “nominators,” then vaccinate one randomly chosen neighbor of each nominator. Ensure all selected neighbors are unique (no duplicates).
- vi) Now we can use the EoN package to simulate disease spread. Install the package in your notebook using: `%pip install EoN`. You will make use of the `EoN.fast_SIR()` function to simulate each realization. This function is just a more efficient and faster version of the stochastic SIR model you all made.

You can use the attached python notebook to see a sample simulation with 10 realizations and no vaccinations. This would help you understand how the function works and what you will need to plot.

After running your simulation, you have will have to plot the number of recovered people for each strategy (look at the sample notebook to understand what I mean) vs the proportion of vaccinated population.

Using this plot answer the following questions:

- a) Which vaccination strategy was most effective?
- b) If you were the policymaker, which strategy would you adopt and why? Consider cost, data availability, and feasibility.
- c) If one vaccination strategy outperformed the others, explain why. What features of the network structure or selection method made that strategy more effective?
- d) How do you expect each vaccination strategy to perform if we simulated the epidemic on an Erdos–Renyi network? Justify your logic and include the same figure as before for this network.